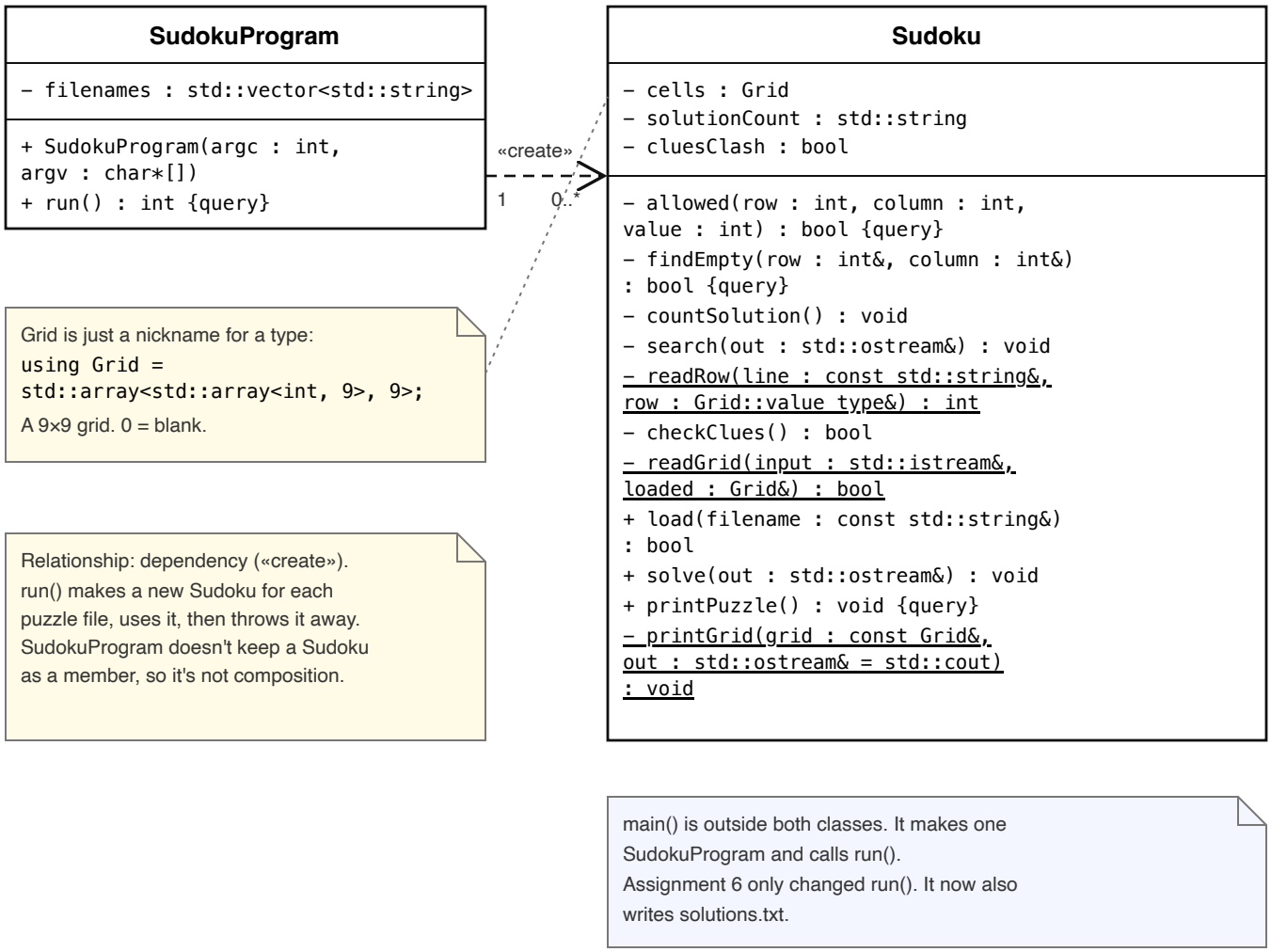
EECS 348 Assignment 6: UML Class Diagrams

Francis Tryban · Assignment6.cpp (modified from Luke Tidball's Assignment 5) · November 13, 2026
Made with Claude Opus 5.5 (through Claude Code) from my final Assignment6.cpp.

Key: − = private, + = public. Underlined = static. `{query}` = const (doesn't change the object).



Class: SudokuProgram

Member	Visibility	Kind	Purpose
filenames : std::vector<std::string>	private	attribute	The puzzle files to solve.
SudokuProgram(argc : int, argv : char*[])	public	constructor	Uses the file names you type, or puzzle1–5.txt.
run() : int {query}	public	method	Solves each puzzle and writes it to solutionN.txt and solutions.txt.

Class: Sudoku

Member	Visibility	Kind	Purpose
cells : Grid	private	attribute	The 9×9 puzzle.
solutionCount : std::string	private	attribute	How many solutions it found.
cluesClash : bool	private	attribute	True if the starting numbers break a rule.
allowed(row, column, value : int) : bool {query}	private	method	Can this number go here?
findEmpty(row : int&, column : int&) : bool {query}	private	method	Finds the next blank.
countSolution() : void	private	method	Adds 1 to the count.
search(out : std::ostream&) : void	private	method	Tries numbers, backs up when stuck, writes each solution.
readRow(line : const std::string&, row : Grid::value_type&) : int	private, static	method	Reads one row.
checkClues() : bool	private	method	Checks the starting numbers.
readGrid(input : std::istream&, loaded : Grid&) : bool	private, static	method	Reads all nine rows.
load(filename : const std::string&) : bool	public	method	Loads a puzzle file.
solve(out : std::ostream&) : void	public	method	Solves it and writes the count or "No solution found."
printPuzzle() : void {query}	public	method	Prints the puzzle. Left over from Assignment 4, not used.
printGrid(grid : const Grid&, out : std::ostream& = std::cout) : void	private, static	method	Writes a grid the same way the input looks.